Maulana Abul Kalam Azad University of Technology, West Bengal

(Formerly West Bengal University of Technology)

Syllabus for B. Tech in Civil & Environmental Engineering

(Applicable from the academic session 2018-2019)

Curriculum Structure

SECOND YEAR - THIRD SEMESTER							
THEORY							
Sl. No.	Code	Theory	Contact hours/wk				Credit point
			L	T	P	Total	
1.	HU301	Values and Ethics in Profession	1	0	0	1	1
2.	CH(CHE)302	Chemistry-2	3	2	0	5	1.5
3.	CH(FT) 301	Basic Environmental Engineering and Elementary Biology	3	0	0	3	1.5
4.	FT301	Thermodynamics & Kinetics	2	0	0	2	2
5.	CE302	Surveying	2	1	0	3	2
6.	CE303	Building Material & Construction	3	2	0	5	1.5
Total of Theory						19	14
PRACTICAL							
7.	CH391	Environmental Engineering Lab	0	0	3	3	2
8.	CH392	Chemistry-2 Lab	0	0	3	3	2
9.	CE392	Surveying Practice-I	0	0	3	3	2
10.	CE393	Building Design &Drawing	0	0	3	3	2
Total of Practical						12	08
Total of Semester						**31**	**17.5**

SECOND YEAR - FOURTH SEMESTER							
THEORY							
Sl. No.	Code	Theory	Contact hours/wk				Credit point
			L	T	P	Total	
1.	M(CS)401	Numerical Methods	2	1	0	3	2
2	CHE414	Unit Operation of chemical Engineering-I	2	0	0	2	2
3.	M402	Mathematics-3	2	1	0	3	2
4	CE402A	Structural Analysis-A	2	1	0	3	2
5.	CE402B	Structural Analysis-B	2	1	0	3	2
6.	CE403A	Soil mechanics-A	2	1	0	3	2
7.	CE403B	Soil mechanics-B	2	1	0	3	2
Total of Theory						20	14
PRACTICAL							
8.	HU481	Tech. Writing/Lang. Lab	0	0	3	3	2
9.	M(CS)491	Numerical Methods	0	0	2	2	1
10.	CE492	Surveying Practice - II	0	0	3	3	2
11.	CE493	Soil Mechanics Lab-I	0	0	3	3	2
12.	CHE484	Unit Operation Lab -I	0	0	3	3	2
Total of Practical						14	9
Total of Semester						**34**	**23**

Maulana Abul Kalam Azad University of Technology, West Bengal

(Formerly West Bengal University of Technology)

Syllabus for B. Tech in Civil & Environmental Engineering

(Applicable from the academic session 2018-2019)

THIRD YEAR - FIFTH SEMESTER							
THEORY							
Sl. No.	Code	Theory	Contact hours/wk				Credit point
			L	T	P	Total	
1.	HU501	Economics for Engineers	1	0	0	1	1
2	CHE514	Unit Operation of Chemical Engineering-II	2	1	0	3	2
3.	CE502A	Design of RC Structures-A	2	1	0	3	2
4.	CE502B	Design of RC Structures-B	2	1	0	3	2
5.	CE503A	Concrete Technology-A	2	0	0	2	2
6.	CE503B	Concrete Technology-B	2	0	0	2	2
7.	CE504A	Engineering Geology-A	2	0	0	2	2
8.	CE504B	Engineering Geology-B	2	0	0	2	2
Total of Theory						18	15
PRACTICAL							
6.	CE591	Soil Mechanics Lab-II	0	0	3	3	2
7.	CE592	Concrete Lab	0	0	3	3	2
8.	CE593	Quantity Surveying , Specification and Valuation	0	0	3	3	2
9.	CE594	Engineering Geology Laboratory	0	0	3	3	2
Total of Practical						12	8
Total of Semester						**30**	**23**

THIRD YEAR - SIXTH SEMESTER							
THEORY							
Sl. No.	Code	Theory	Contact hours/wk				Credit point
			L	T	P	Total	
1.	HU601	Principles of Management	2	0	0	2	2
2	FT604	Microbial Tech & Biotechnology	3	0	0	3	3
3.	CE602	Design of Steel Structure	3	0	0	3	3
4.	CE603	Construction Planning& Management	3	0	0	3	3
5	CVE601	Professional Elective-I	3	0	0	3	3
6.	CVE602	Free Elective- I	3	0	0	3	3
Total of Theory						17	17
PRACTICAL							
7.	CVE 691	Environmental Chemistry Lab	0	0	3	3	2
8.	FT692	Microbial Tech Lab	0	0	4	4	2
9.	CVE681	Seminar	0	0	3	3	2
Total of Practical						10	6
Total of Semester						27	23

Professional Elective-I

1. CVE601A: Solid & Hazardous Waste Treatment
2. CVE601B: Industrial Waste Treatment

Free Elective –I

1. CVE 602A : Operations Research (M)
2. CVE 602B : Human Resource Management (HSS)
3. CVE 602C : Materials Handling (ME)

FOURTH YEAR - SEVENTH SEMESTER							
THEORY							
Sl. No.	Code	Theory	Contact hours/wk				Credit point
			L	T	P	Total	
1.	CE701	Environmental Engineering	2	0	0	2	2
2	CE702	Water Resource Engineering	2	0	0	2	2
3.	CVE701	Professional Elective-II	1	0	0	1	1
4.	CVE702	Professional Elective-III	1	0	0	1	1
5.	CVE703	Free Elective- II	1	0	0	1	1
Total of Theory						7	7
PRACTICAL							
6.	CVE791	Environmental Engineering & Design Lab	0	0	3	3	2
7.	CVE792	Free Elective Laboratory	0	0	3	3	2
8.	CVE781	Industrial Training	4 weeks duration during 6th -7th Semester break				2
9.	CVE782	Project Part I				6	2
Total of Practical						12	8
Total of Semester						19	15

Professional Elective –II

1. CVE701A: Soil Stabilization and Ground Improvement Techniques
2. CVE701B: Advanced Highway and Transportation Engineering

Profession Elective –III

1. CVE702A: Advanced Structural Analysis
2. CVE702B: Hydraulic Structures

3. CVE702C: Climate & Ocean Currents

Free Elective –II

1. CVE703A : Engineering Materials (ME)
2. CVE703B : Electrical and Electronic

Measurement (EE) **Free Elective Laboratory**

1. CVE792A : Material Testing Lab (ME)
2. CVE792B : Electrical and Electronic Measurement Laboratory (EE)
3. CVE792C: Climate & Ocean Currents Lab

FOURTH YEAR – EIGHTH SEMESTER

THEORY							
Sl. No.	Code	Theory	Contact hours/wk				Credit point
			L	T	P	Total	
1.	HU801A HU801B	Organizational Behavior/Project Management	2	0	0	2	2
2	CVE801	Professional Elective-IV	2	0	0	2	2
3.	CVE802	Professional Elective-V	2	0	0	2	2
4.	CVE803	Natural Resources : Renewable and Non-renewable Resources	2	0	0	2	2
5.	CVE804	Cost Effective Effluent Treatment	2	0	0	2	2
Total of Theory						10	10
PRACTICAL							
5.	CVE881	Project Part II	0	0	12	12	4
6.	CVE882	Grand Viva					2
Total of Practical						12	6
Total of Semester						22	16

Professional Elective –IV

1. CVE801A: Environment Pollution and Control (CE)
2. CVE801B: Water Resource Management and Planning (CE)
3. CVE801C: Remote Sensing and

GIS (CE) **Professional Elective –V**

1. CVE802A: Finite Element Method (CE)
2. CVE802B: Pavement Design (CE)